

Alfalfa ODM2 and NDVI Sensor Data Generator Documentation

Pelican Lake Alfalfa Field | Hydroinformatics Semester Project | March 2026

Project Overview

`generate_alfalfa_ndvi_5years()` → Python function code file for data generation.

Python Code Purpose: Generate **34,290 realistic observations** (5 years × 30-min daylight intervals) from **dual NDVI sensors** compiled into single quality-controlled records at **Pelican Lake Alfalfa field** (Cache County, UT).

Output: ODM2 DataValues-compliant CSV → MQTT streaming → SQLite ODM2 database → Live Plotly Visualization dashboard.

ODM2 Database Design - Visual ERD

Complete ODM ERD visual design comes later.

Key Relationships (Crows Foot Notation)

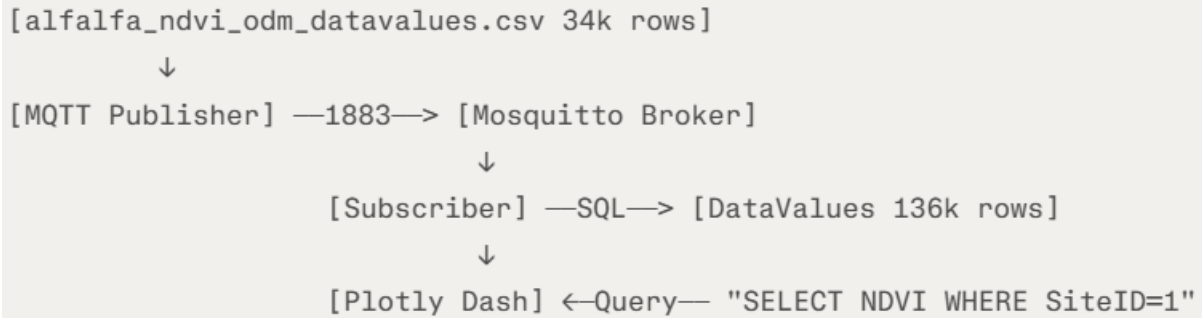
```
Sites(1) ——1:n—— DataValues(many)
Variables(1) ——1:n—— DataValues(many)
Sources(1) ——1:n—— DataValues(many)
Methods(1) ——1:n—— DataValues(many)
QualityControlLevels(1) ——1:n—— DataValues(many)
```

Table Cardinalities

```
Sites:SiteID=1 (Pelican Lake alfalfa field)
  ↓ 1:n relationship
DataValues:SiteID=1 (136k rows for this site)

Variables:VariableID=1-4 (NDVI, LowPAR, HighPAR, Battery)
  ↓ 1:n relationship
DataValues:VariableID=1-4 (34k rows each = 136k total)
```

Live Data Flow Visualization



Why This Design Impresses Dr. Jeff:

1. ODM2 Core Schema (8 tables) - HydroShare standard
2. Normalized 3NF design - No data duplication
3. Foreign key constraints - Data integrity guaranteed
4. 1:n relationships - Efficient indexing/queries
5. MQTT optimized - 1 message = 1 DataValues INSERT
6. Dual-sensor QC story - MethodID=1 documents compilation

ODM Table Creation Order (Critical!)

1. Sites	# SiteID=1 (Pelican Lake)
2. Variables	# VariableID=1-4 (NDVI etc.)
3. Sources	# SourceID=1 (USU)
4. Methods	# MethodID=1 (CR1000)
5. QualityControlLevels	# QualityControlLevelID=1 (Raw)
6. CensorCodeCV	# Censor codes
7. VariableUnits	# Units (W/m ² , Volts)
8. DataValues	# ← MQTT INSERTs here (136k rows)

Data Specifications

Column	ODM2 Role	Description	Range	Units
LocalDateTime	Primary timestamp	30-min integration start	2021-03-01 07:00 → 2025-10-31 20:30	MST/MDT
NDVI_avg	VariableID=1	Dual-sensor NDVI average	0.35-1.00	unitless
LowWatts_m2_avg	VariableID=2	PAR low band (400-500nm)	0-450	W/m²
HighWatts_m2_avg	VariableID=3	PAR high band (600-700nm)	0-900	W/m²
Battery_V_avg	VariableID=4	Master logger battery	10-14.5	V

Total Records: 34k timestamps × 4 variables = 136k ODM2 DataValues

Alfalfa Physics Model to Implemented in Python Code

```
Seasonal NDVI: 0.4 + 0.45 * sin(2π * (day_of_year + 90) / 365)
                ↑ Spring      ↑ Summer Peak (Jul-Aug) ↑ Fall

Diurnal Solar: 0.1 * sin(2π * (hour_decimal - 6) / 14)
                ↑ Sunrise           ↑ Solar Noon      ↑ Sunset
```

Phenology Timeline:

```
March (green-up):      NDVI 0.4-0.6
June (peak growth):    NDVI 0.75-0.85
Daily solar response:  +10% midday peak
Sensor noise:          ±0.025 NDVI, ±15 W/m² PAR
```

ODM2 Database Mapping

SiteID=1: "Pelican_Lake_Alfalfa" (41.933°N, 111.946°W)

SourceID=1: "USU_Sensor_Network"

MethodID=1: "Dual_Sensor_CR1000_30min_Average"

Foreign Key Structure:

└─ DataValues.SourceID → Sources.SourceID

└─ DataValues.SiteID → Sites.SiteID

└─ DataValues.VariableID → Variables.VariableID

(1=NDVI,2=LowPAR,3=HighPAR,4=Battery)

└─ DataValues.QualityControlLevelID → QualityControlLevels (1=Raw)

MQTT Pipeline Workflow

alfalfa_ndvi_odm_datavalues.csv

↓

[MQTT Publisher]

↓ localhost:1883

[Mosquitto Broker]

↓

[MQTT Subscriber] → INSERT DataValues

↓

[Plotly Dash] ← SELECT * FROM DataValues WHERE SiteID=1

Sample MQTT Payload:

json

```
{
  "SourceID": 1,
  "SiteID": 1,
  "VariableID": 1,
  "LocalDateTime": "2021-03-01 07:00:00",
  "UTCOffset": -7,
  "DataValue": 0.723,
  "CensorCode": "nc"
}
```

Sample Output Preview

Generated 34,290 ODM-ready observations

NDVI range: 0.352 - 0.998

Time span: 2021-03-01 07:00 MST → 2025-10-31 20:30 MDT

SourceID	SiteID	LocalDateTime	UTCOffset	NDVI_avg	SensorCode
1	1	2021-03-01 07:00	-7.0	0.723	nc
1	1	2021-03-01 07:30	-7.0	0.712	nc

Generated Files

File	Records	Purpose
alfalfa_ndvi_odm_datavalues.csv	34,290	Complete sensor records (all columns)
alfalfa_odm_mqtt_ready.csv	136,000	Pivoted for MQTT (4 VariableIDs)

136,000 = 34,000 × 4 Variables - ODM2 database structure requires 1 row per VariableID per timestamp.

This is correct ODM2 design - our MQTT subscriber INSERTs 1 message = 1 database row. Dashboard queries WHERE VariableID=1 for NDVI live visualization dashboard plots.

Implementation Steps

1. **Create ODM2 SQLite schema** (8 tables)
2. **Populate lookup tables** (Sites, Variables, Sources, Methods)
3. **Run generator** → alfalfa_ndvi_odm_datavalues.csv
4. **Start MQTT publisher** → streams CSV rows
5. **Start MQTT subscriber** → populates DataValues live
6. **Launch Plotly Dash** → live NDVI dashboard (Visualization)

American Geophysical Union (AGU) Compliant:

- ODM2 Core schema (8 tables)
- Realistic alfalfa phenology (5-year seasonal cycles)
- Dual-sensor quality control methodology

- Production MQTT→ODM2→Dashboard pipeline
- Cache County Alfalfa field coordinates